

EVENING

[Total No. of Questions: 09]

04 JUN 2024

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Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Object Oriented Programming using C++

Subject Code: PCIT-102

Paper ID: 16041

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) How do you declare and initialize an array in C++?
- b) How are keywords different from identifiers in C++? Give examples of keywords.
- c) Differentiate between unary and binary operator overloading.
- d) How will you declare and define a constructor?
- e) Compare the object-oriented paradigm with the procedural programming paradigm.
- f) Write the C++ syntax to open a file named "data.txt" in write mode.

Part – B

[Marks: 04 each]

- Q2.** Differentiate between call by value and call by reference.
- Q3.** Explain in detail the concept of try, catch, and throw in C++ exception handling.
- Q4.** Can a class have both data members and member functions? Explain in detail.
- Q5.** Write a C++ program to demonstrate the use of the scope resolution operator by defining the member functions of a **Student** class outside the class.
- Q6.** Write a C++ program that uses a for loop to iterate through an array of integers and prints each element.
- Q7.** Develop a program in C++ for finding the length of a string and copying one string into the other.

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Part – C

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[Marks: 12 each]

- Q8. Explain the steps involved in opening and closing a file in C++, highlighting the importance of proper file handling. Write a C++ function that reads data from a file named "input.txt" and displays it on the console.

OR

Illustrate the concept of abstract classes. Why there is a need of pointers? Also explain the use of friend function and compare it with virtual functions.

- Q9. How does the continuous statement is different from the break statement in C++. Develop a program that simulates a simple calculator with options for addition, subtraction, multiplication and division using switch statement.

OR

Create a class to represent a bank account, include the following members:

a. Data members:

- i. Name of the depositor
- ii. Account number

2. Member functions

- i. Deposit an amount
- ii. Withdraw an amount
- iii. Display name and balance

Write a program to print full pyramid using characters.
